

Algebra: Adding & Subtracting Rational Expressions

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DIRECTIONS

Find the LCD, rewrite each fraction, then add or subtract. Simplify fully.

1 Add (different denominators):

$$\frac{3}{4x} + \frac{4}{7x}$$

Answer: _____

2 Add (same denominator):

$$\frac{1}{x+5} + \frac{5}{x+5}$$

Answer: _____

3 Add (different denominators):

$$\frac{4}{3x} + \frac{1}{6x}$$

Answer: _____

4 Add (same denominator):

$$\frac{3}{x+5} + \frac{4}{x+5}$$

Answer: _____

5 Add (same denominator):

$$\frac{5}{x+3} + \frac{4}{x+3}$$

Answer: _____

6 Add (same denominator):

$$\frac{3}{x+6} + \frac{3}{x+6}$$

Answer: _____

7 Subtract (binomial denominators):

$$\frac{4}{x+2} - \frac{3}{x+2}$$

Answer: _____

8 Subtract (binomial denominators):

$$\frac{4}{x+5} - \frac{2}{x+5}$$

Answer: _____

9 Subtract (binomial denominators):

$$\frac{4}{x+1} - \frac{3}{x+4}$$

Answer: _____

10 Add (different denominators):

$$\frac{2}{4x} + \frac{4}{5x}$$

Answer: _____



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Answer Key & Solutions

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TEACHER NOTES Factor all denominators first to find the LCD. Remind students to watch signs when subtracting numerators.

1 Add (different denominators):

$$= \frac{37}{28x}$$

$$\frac{3}{4x} + \frac{4}{7x}$$

2 Add (same denominator):

$$= \frac{6}{x+5}$$

$$\frac{1}{x+5} + \frac{5}{x+5}$$

3 Add (different denominators):

$$= \frac{3}{2x}$$

$$\frac{4}{3x} + \frac{1}{6x}$$

4 Add (same denominator):

$$= \frac{7}{x+5}$$

$$\frac{3}{x+5} + \frac{4}{x+5}$$

5 Add (same denominator):

$$= \frac{9}{x+3}$$

$$\frac{5}{x+3} + \frac{4}{x+3}$$

6 Add (same denominator):

$$= \frac{6}{x+6}$$

$$\frac{3}{x+6} + \frac{3}{x+6}$$

7 Subtract (binomial denominators):

$$= \frac{4(x+2) - 3(x+2)}{(x+2)(x+2)}$$

$$\frac{4}{x+2} - \frac{3}{x+2}$$

8 Subtract (binomial denominators):

$$= \frac{4(x+5) - 2(x+5)}{(x+5)(x+5)}$$

$$\frac{4}{x+5} - \frac{2}{x+5}$$

9 Subtract (binomial denominators):

$$= \frac{4(x+4) - 3(x+1)}{(x+1)(x+4)}$$

$$\frac{4}{x+1} - \frac{3}{x+4}$$

10 Add (different denominators):

$$= \frac{13}{10x}$$

$$\frac{2}{4x} + \frac{4}{5x}$$



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